



# Comparing the pyrolysis kinetics of dimethoxymethane and 1,2-dimethoxyethane: An experimental and kinetic modeling study

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## ARTICLE INFO

### Article history:

Received 15 July 2020

Revised 6 December 2020

Accepted 7 December 2020

Available online 24 December 2020

### Keywords:

Dimethoxymethane (DMM)

1,2-Dimethoxyethane (1,2-DME)

Pyrolysis

Kinetic modeling

Polyether fuels

## ABSTRACT

This work investigates the reaction kinetics of the thermal decomposition of dimethoxymethane (DMM) and 1,2-dimethoxyethane (1,2-DME), which are both representative molecules of the promising clean polyether fuels. Pyrolysis experiments are carried out with helium diluted mixtures containing individual fuels in a flow tube at two different pressures of 760 Torr and 30 Torr. By varying the temperature distributions along the flow tube, the resulting species composition is sampled downstream the reactor and analyzed by a photoionization mass spectrometer. A kinetic model, including sub-mechanisms for the thermal decomposition of both fuels, is proposed, which can well characterize important measurements, such as the fuel decomposition reactivity and the speciation of crucial products. Under identical conditions, 1,2-DME exhibits a higher decomposition reactivity than DMM, because of the characteristic C–C bond fission as well as the easier hydrogen abstraction reactions by methyl radical. Some fuel-specific intermediates are observed, including methyl formate in DMM pyrolysis, methyl vinyl ether and methoxy acetaldehyde in 1,2-DME pyrolysis. These species mainly come from fuel radical dissociations following the hydrogen abstractions from the central (–CH<sub>2</sub>O–) and (–CH<sub>2</sub>CH<sub>2</sub>O–) moieties in DMM and 1,2-DME, respectively. The formation of some other intermediates are closely related to the consumption of these fuel-specific species. Particularly, the decomposition of methyl formate produces high concentrations of methanol in DMM pyrolysis. In 1,2-DME pyrolysis, the consumption of methyl vinyl ether and methoxy acetaldehyde brings about the formation of C<sub>2</sub> oxygenated intermediates such as acetaldehyde, ethenol and ketene. Higher concentrations of unsaturated C<sub>2</sub> (ethylene and acetylene) and larger hydrocarbon intermediates present in 1,2-DME pyrolysis, due to the existence of C–C bond in the fuel molecule.

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## 1. Introduction

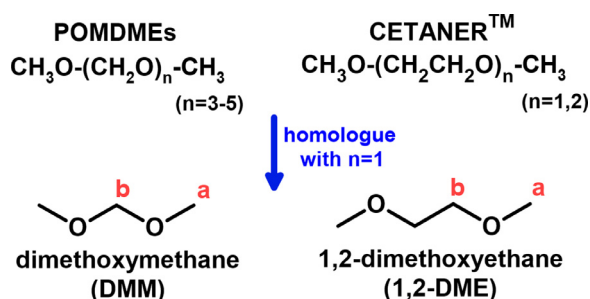
Progress is being made in the search for clean alternative fuels to better regulate emissions from combustion processes [1]. Diverse oxygenated fuels or fuel additives have been applied in practical engine operations, and some of them show superiorities from different aspects, such as good physical-chemical properties and the abilities in reducing emission and improving ignition. Poly(oxyethylene) dimethyl ethers (POMDMes) are among the most attractive fuel additives, appealing large research interest,

thanks to the compatibility with existing diesel engines and the outstanding soot-reducing ability [2,3]. CETANER<sup>TM</sup> is a commercial fuel additive that effectively reduces emissions, and meanwhile its glycol ether compositions can improve the ignition properties of diesel fuels [4,5]. As displayed in Scheme 1, POMDMes and CETANER<sup>TM</sup> are polyether mixtures with the uniform chemical formulas of CH<sub>3</sub>O(CH<sub>2</sub>O)<sub>n</sub>CH<sub>3</sub> and CH<sub>3</sub>(CH<sub>2</sub>CH<sub>2</sub>O)<sub>n</sub>CH<sub>3</sub>, respectively. The application of these fuels requires in-depth knowledge of the detailed reaction networks that convert fuels into final products via a variety of intermediates. Such information is normally interpreted by combustion kinetic models, which are demanded in the numerical design of advanced combustion devices. Scheme 1 also presents the simple homologues with *n* = 1, dimethoxymethane (DMM) and 1,2-dimethoxymethane (1,2-DME). They contain both the (CH<sub>3</sub>O–) ends and the central (–CH<sub>2</sub>O–) or (–CH<sub>2</sub>CH<sub>2</sub>O–) unit,

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**Scheme 1.** Chemical formulas for the commercial polyether fuels POMDME and CETANER™, and the molecular structures for the homologues with  $n = 1$ , DMM and 1,2-DME. Fuel radicals formed by losing H atoms from the terminal and middle carbon atoms are named as a-type and b-type fuel radicals, respectively.

so that can well represent the chemical structures of POMDMEs and CETANER™, respectively. The pursuit of the combustion kinetics of DMM and 1,2-DME potentially offers insights into larger homologues that compose the practical alternative fuels, for instance, how the terminal and central moieties separately contribute to species formation under combustion conditions. Undoubtedly, predictive kinetic models for DMM and 1,2-DME combustion lay essential foundations for the development of combustion kinetic models of POMDMEs and CETANER™.

The “HyChem” concept [6] was recently proposed in modeling the combustion of practical fuels. The essential idea is to decouple the sophisticated combustion processes into a concise reaction scheme of fuel pyrolysis and a detailed chemical mechanism describing the subsequent oxidation procedure. Following the principle of this strategy, it is a prerequisite step to identify the predominant pathways leading to fuel thermal decomposition. For both DMM and 1,2-DME, most previous relevant studies focused on the combustion kinetics under flame [7–9] and low-temperature oxidation [10–13] conditions, while the fuel thermal decomposition pathways are not as clear yet, since relevant reactions, such as unimolecular decompositions, are not sensitive under oxidative conditions. DMM thermal decomposition was investigated under shock tube conditions by several groups recently [14–16], but the dominant channel is still disputed [14,16] between the two type of C–O cleavages, leading to  $\text{CH}_3\text{OCH}_2\dot{\text{O}}+\text{CH}_3$  and  $\text{CH}_3\text{O}\dot{\text{C}}\text{H}_2+\text{CH}_3\text{O}$ , respectively. With the later pathway being taken as the major one, a kinetic model by Peukert et al. [14] could well reproduce their shock tube experimental results; while the recent theoretical work by Golka et al. [15] indicated that the former channel governs the decomposition of DMM. The pyrolysis kinetics of 1,2-DME has not been addressed in literature to the best of our knowledge. A good understanding of the pyrolysis kinetics of DMM and 1,2-DME is urged by the construction of predictable combustion kinetic models for the polyether alternative fuels.

The design of high-performance fuels or fuel mixtures is based on the fuel diversity and the fact that fuel molecular structures are influential on their combustion properties [17]. The currently studied DMM and 1,2-DME share a few structural similarities: they both have symmetrical chained structures containing double ether functionalities, two ( $-\text{CH}_3\text{O}$ ) terminal groups as well as two type of C–H bonds. These common characteristics may result in similar reaction pathways in their thermal decomposition schemes. On the other hand, DMM and 1,2-DME also have distinct molecular traits in their central ( $-\text{CH}_2\text{O}-$ ) and ( $-\text{CH}_2\text{CH}_2\text{O}-$ ) units, which potentially differs their speciation behaviors. Revealing potential influences of the mentioned fuel structural features on their pyrolysis kinetics can be beneficial to the future fuel design.

Inspired by the above necessities, a comparative study on DMM and 1,2-DME pyrolysis is carried out via experimental and kinetic

modeling methods. Flow reactor pyrolysis experiments of both fuels are conducted at two pressure conditions, and the consequent species mole fractions that evolve with the reacting temperatures are probed through the molecular beam mass spectrometry technique. A kinetic model is proposed by sorting and comparing potential common and fuel-specific consumption channels of DMM and 1,2-DME decomposition. The aim of the kinetic model is not only to satisfactorily predict the speciation measurements, but also to interpret the similarities and differences in the reaction schemes of DMM and 1,2-DME thermal decomposition. Particular attention is paid to the influences of the common and distinct fuel structural features.

## 2. Flow reactor pyrolysis experiments

Experimental measurements are carried out with the flow reactor pyrolysis facility at National Synchrotron Radiation Laboratory (NSRL) in Hefei, China. The details about the beamline and the experimental apparatus can be found in previous publications [18,19]. Briefly, the experimental set-up consists of three parts, a flow tube reactor, a molecular beam sampling system and a photoionization mass spectrometer. A boron nitride tube (inner diameter 11 mm), evenly wound with heating wires, is mounted in the center of a homemade furnace. It coaxially holds an  $\alpha$ -type alumina ceramic tube, which is 220 mm in length and 7 mm in inner diameter. A tungsten-rhenium (W-Re) thermocouple is positioned near the center of the heating zone outside the heating wires to monitor and control the temperature distribution along the flow-tube. During the experiments, a helium (He) diluted gas mixture, containing 1% fuel (DMM or 1,2-DME) and 1% krypton (Kr) as the internal calibrant, is fed into the ceramic tube at a constant volume flow rate of 1000 standard cubic centimeters per minute (SCCM). The chosen total flow rate is consistent with those adopted in previous studies on fuel pyrolysis using the same setup [20–25]. It is proven that the current experimental configuration satisfies the criterions of a large Péclet number (Pe) and a small vessel dispersion number ( $D_{\text{disp}}/\text{u}D$ ) [23]. Diffusion in the axial direction of the flow reactor is negligible so that it can be modeled with a “plug-flow reactor”. The liquid samples (DMM and 1,2-DME, with purities of over 99%, purchased from Sigma-Aldrich) are metered by a high-pressure chromatography pump into a vaporizer, and the gas flows of Kr and He are separately regulated by MKS mass flow controllers. The current experiments are performed under two different pressure conditions, by controlling the pressure at the reactor exit at 760 Torr and 30 Torr, respectively, and it is proven that the pressure can be treated as constant along the tube [19,20]. The temperature distributions are changed by varying the heating power, and in each experiment, after the temperature conditions stabilize, the gas mixture is sampled downstream the tube with a quartz nozzle. Nozzles with orifice diameters of 100  $\mu\text{m}$  and 500  $\mu\text{m}$  are used for experiments at 760 Torr and 30 Torr, respectively, for a purpose of obtaining enough signal sensitivity while keeping the ionization chamber pressure within a safe range. A molecular beam, formed by the sampled gas, is crossed by the synchrotron radiation, and the resulting ions are detected by a time-of-flight mass spectrometer. Two operation modes are employed in the experiments: scanning the photon energy under fixed temperature conditions to obtain the photoionization efficiency (PIE) spectra for species identification; recording the mass spectra at specific photon energies (9.5, 10.0, 10.5, 11.0, 11.5, 12.3, 13.2, 14.35 and 15.36 eV) under individual temperature conditions to derive the species mole fractions.

The data processing for the quantification is based on the following equation:

$$S_i(T, E) \propto X_i(T) \times \sigma_i(E) \times D_i \times \lambda(T) \quad (1)$$

where for species  $i$ , the measured signal  $S_i(T, E)$  is dependent on the temperature ( $T$ ) and the photon energy ( $E$ ), and it is proportional to the mole fraction  $X_i(T)$ , the photoionization cross section at the photon energy  $E$ ,  $\sigma_i(E)$ , the mass discrimination factor  $D_i$  and the gas expansion factor  $\lambda(T)$ .  $\lambda(T)$  can be expressed by the normalized Kr signal at 15.36 eV:

$$\lambda(T) = S_{Kr}(T, 15.36 \text{ eV}) / S_{Kr}(T_0, 15.36 \text{ eV}), \quad (2)$$

in which  $T_0$  is the initial temperature before the fuel starts to decompose. For the fuel, the initial mole fraction  $X_F(T_0)$  is known to be 1%, therefore the fuel mole fraction as a function of the temperature can be derived from Eq. (3):

$$X_F(T) = \frac{S_F(T, E)}{S_F(T_0, E)} \times X_F(T_0) \times \lambda(T) \quad (3)$$

Mole fractions of the fuel are then used as a reference to calculate those for other species,  $X_i(T)$ , according to Eq. (4):

$$\frac{X_i(T)}{X_F(T)} = \frac{S_i(T, E)}{S_F(T, E)} \times \frac{\sigma_F(E)}{\sigma_i(E)} \times \frac{D_F}{D_i} \quad (4)$$

The uncertainty in the measured species mole fractions largely depends on the accuracy of the used photoionization cross section (PICS) data. Measured PICSs are available for most species detected in the current experiments, and the uncertainty factors range from 1.3 to 1.7, by considering the stability of the products and whether there are isomers to be resolved. Methyl radical mole fractions are measured in the experiments at 30 Torr, for which the uncertainty is estimated to be a factor of 2 because of the potential quenching effects from the sampling nozzle, though measured PICS data are available. Methyl vinyl ether ( $\text{CH}_3\text{OC}_2\text{H}_5$ ) and methoxy-acetaldehyde ( $\text{CH}_3\text{OCH}_2\text{CHO}$ ) are identified as important intermediates in 1,2-DME pyrolysis experiments through PIE measurements, as shown in Fig. S1 in the *Supplemental Material*. However, measured PICS data are not available for these two species, and therefore the estimated PICSs are used in the quantification, bringing an uncertainty factor of 2 in the mole fractions. The used PICSs and mass discrimination factors for individual species are provided in Table S1 in the *Supplementary Material* together with the evaluated uncertainty factors in the mole fraction measurements.

The species composition probed at the exit of the flow reactor evolves with the temperature conditions inside the reactor, which are key parameters in the current experiments. An S-type thermocouple is used to measure the temperature distribution along the flow tube at each experimental condition, with 1000 SCCM He fed into the flow tube. The uncertainty in the measured temperatures is estimated to be within  $\pm 30$  K. Each temperature distribution is denoted by its maximum value ( $T_{\max}$ ), and the measured species mole fractions are shown as a function of  $T_{\max}$  in this work. Technically, it is questionable to directly compare data points at different  $T_{\max}$ s, because the residence time varies with the temperature conditions in the reactor. According to the simulated results, the residence time alters from  $1.99 \times 10^{-1}$  to  $1.28 \times 10^{-1}$  s (over the  $T_{\max}$  range of 783–1166 K) and from  $6.60 \times 10^{-3}$  to  $4.22 \times 10^{-3}$  s (over the  $T_{\max}$  range of 903–1396 K) in the experiments at 760 Torr and 30 Torr, respectively. Nevertheless, the quantitative speciation data can still be used to inspect the predictive performances of the kinetic model under relevant conditions. Moreover, the comparisons between the pyrolysis of the two polyethers under identical conditions can offer insights into their common and fuel-specific reaction kinetics, which is the focus of the current work. Measured species mole fractions and temperature distributions in all experiments are provided in the *Supplementary Material*.

### 3. Kinetic modeling

The recent AramcoMech 3.0 [26] is taken as the core mechanism of the kinetic model for DMM and 1,2-DME pyrolysis. It already includes the reactions of most important oxygenated and hydrocarbon intermediates and products. Fuel consumption sub-mechanisms of DMM and 1,2-DME are developed in the current work. DMM and 1,2-DME are alike in the sense that both molecules contain two oxygen atoms, and this structural characteristic is not seen in other regular ether fuels such as dimethyl ether (DME) and diethyl ether (DEE). The kinetic model construction is partly based on comparisons of potential decomposition pathways between the two polyethers. The assignments of specific kinetic parameters are also facilitated by theoretical calculations. In this section, the used methods for theoretical calculations will be introduced first, and relevant results will be presented in the following descriptions organized by reaction categories. Reactions that are updated and added in the current work are listed in Table S1 together with their rate coefficients.

#### 3.1. Theoretical methods

Theoretical efforts in the present work include two aspects: 1). The tentative exploration of the molecular elimination reaction channels of DMM: The geometries of all involved structures are optimized with the B3LYP/aug-cc-pVTZ method using the Gaussian 09 program [32], and only the lowest energy conformers are considered. The higher level energy estimates are obtained at the CCSD(T)-F12/aug-cc-pVTZ level of theory using MOLPRO program package [33]. 2). The derivation of rate coefficients for hydrogen abstraction reactions from 1,2-DME by H and  $\text{CH}_3$  and the decomposition/isomerization reactions of 1,2-DME fuel radicals: The reaction potential energy surfaces (PESs) are obtained at CCSD(T)/cc-pV(D + T)Z//B2PLYPD3/6–311++g(d,p) level of theory. Based on the calculated energies and molecular properties, the high-pressure limit rate constants are calculated using the transition state theory (TST) over the temperature range of 500–2000 K. The low-frequency modes that resemble internal torsions around single bonds are treated as hindered rotors with the hindrance potentials computed at the level of B2PLYPD3/6–31+g(d,p), while for other vibrational modes, the rigid rotor harmonic oscillator (RRHO) model is used. Tunneling effects are taken into account via the Eckart model [27].

Specific energy values, including bond dissociation energies (BDEs) and energy barriers of equivalent reactions of DMM and 1,2-DME are listed in Table 1, and relevant details will be discussed later in this section.

#### 3.2. Unimolecular decomposition reactions

It is challenging to accurately determine the rate coefficients for DMM and 1,2-DME unimolecular decomposition reactions, in particular for the competing C–O fission channels. Multi-reference electronic structure methods and advanced variational transition state theories need to be implemented, and the multiple oxygen atoms embedded in the fuel carbon chains add to the difficulty. DMM unimolecular decomposition kinetics was the subject of intense recent research [14–16], where attempts were made to obtain either an overall rate expression, or separate rate coefficients for the two competing C–O fissions, R1 and R2. Peukert et al. [14] derived an overall rate expression through shock tube experiments. They [14] noted that the temporal species concentration profiles and product compositions could be well reproduced by using a branching ratio reported in an earlier kinetic model [13], in which R2 was considered as the predominant channel. Very recently, Golka et al. [16] reported pressure-dependent rate

**Table 1**

Bond dissociation energies (BDEs) at 0 K of equivalent bonds and zero-point energy (ZPE) -corrected energy barriers for similar reactions in the reaction systems of DMM and 1, 2-DME. (Units, kcal/mol).

| DMM                                           |                                                                                                                                         |                                                           | 1,2-DME                                                                                                                                                                 |                   |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|                                               | Corresponding bond or reaction                                                                                                          | Value                                                     | Corresponding bond or reaction                                                                                                                                          | Value             |
| BDE (R'-CH <sub>3</sub> )                     | CH <sub>3</sub> OCH <sub>2</sub> O-CH <sub>3</sub>                                                                                      | 82.1 <sup>a</sup> , 82.3 <sup>b</sup> , 82.2 <sup>c</sup> | CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> O-CH <sub>3</sub>                                                                                                      | 82.0 <sup>d</sup> |
| BDE (R''-OCH <sub>3</sub> )                   | CH <sub>3</sub> OCH <sub>2</sub> -OCH <sub>3</sub>                                                                                      | 88.1 <sup>a</sup> , 86.3 <sup>b</sup> , 87.9 <sup>c</sup> | CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> -OCH <sub>3</sub>                                                                                                      | 84.6 <sup>d</sup> |
| BDE (Ra-H)                                    | CH <sub>3</sub> OCH <sub>2</sub> OCH <sub>2</sub> -H                                                                                    | 95.7 <sup>b</sup> , 95.7 <sup>c</sup> , 95.0 <sup>e</sup> | CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> OCH <sub>2</sub> -H                                                                                                    | 96.1 <sup>d</sup> |
| BDE (Rb-H)                                    | CH <sub>3</sub> OCH(-H)OCH <sub>3</sub>                                                                                                 | 96.0 <sup>b</sup> , 95.8 <sup>c</sup> , 95.3 <sup>e</sup> | CH <sub>3</sub> OCH <sub>2</sub> CH(-H)OCH <sub>3</sub>                                                                                                                 | 94.4 <sup>d</sup> |
| TS (F + H => Ra+H <sub>2</sub> )              | CH <sub>3</sub> OCH <sub>2</sub> OCH <sub>3</sub> +H = CH <sub>3</sub> OCH <sub>2</sub> OCH <sub>2</sub> +H <sub>2</sub>                | 8.2 <sup>b</sup> , 8.4 <sup>e</sup>                       | CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> OCH <sub>3</sub> +H = CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> OCH <sub>2</sub> +H <sub>2</sub>                | 7.8 <sup>e</sup>  |
| TS (F + H => Rb+H <sub>2</sub> )              | CH <sub>3</sub> OCH <sub>2</sub> OCH <sub>3</sub> +H = CH <sub>3</sub> OCHOCH <sub>3</sub> +H <sub>2</sub>                              | 7.6 <sup>b</sup> , 7.7 <sup>e</sup>                       | CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> OCH <sub>3</sub> +H = CH <sub>3</sub> OCH <sub>2</sub> CHOCH <sub>3</sub> +H <sub>2</sub>                              | 6.7 <sup>e</sup>  |
| TS (F+CH <sub>3</sub> => Ra+CH <sub>4</sub> ) | CH <sub>3</sub> OCH <sub>2</sub> OCH <sub>3</sub> +CH <sub>3</sub> = CH <sub>3</sub> OCH <sub>2</sub> OCH <sub>2</sub> +CH <sub>4</sub> | 12.4 <sup>b</sup>                                         | CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> OCH <sub>3</sub> +CH <sub>3</sub> = CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> OCH <sub>2</sub> +CH <sub>4</sub> | 12.4 <sup>e</sup> |
| TS (F+CH <sub>3</sub> => Rb+CH <sub>4</sub> ) | CH <sub>3</sub> OCH <sub>2</sub> OCH <sub>3</sub> +CH <sub>3</sub> = CH <sub>3</sub> OCHOCH <sub>3</sub> +CH <sub>4</sub>               | 12.6 <sup>b</sup>                                         | CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> OCH <sub>3</sub> +CH <sub>3</sub> = CH <sub>3</sub> OCH <sub>2</sub> CHOCH <sub>3</sub> +CH <sub>4</sub>               | 10.9 <sup>e</sup> |
| TS (Ra => CH <sub>2</sub> O+R'')              | CH <sub>3</sub> OCH <sub>2</sub> OCH <sub>2</sub> = CH <sub>3</sub> OCH <sub>2</sub> +CH <sub>2</sub> O                                 | 25.5 <sup>b</sup>                                         | CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> OCH <sub>2</sub> = CH <sub>3</sub> OCH <sub>2</sub> CH <sub>2</sub> +CH <sub>2</sub> O                                 | 23.8 <sup>e</sup> |

<sup>a</sup> CCSD(F12\*)(T\*)/cc-pVQZ-F12//B2PLYP-D3/def2-TZVPP [16].

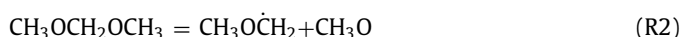
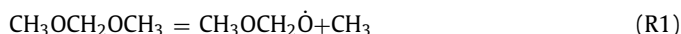
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<sup>c</sup> CCSD(T)-F12/aug-cc-pVTZ//B3LYP/aug-cc-pVTZ, this work.

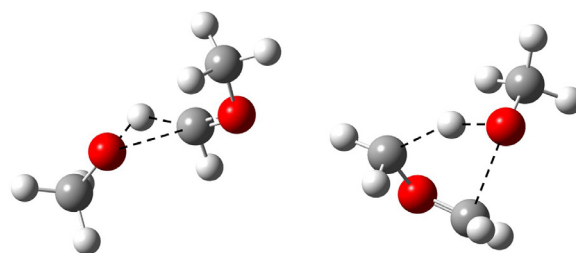
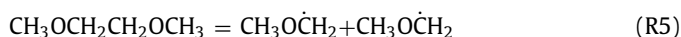
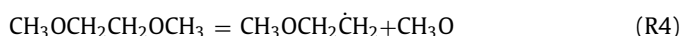
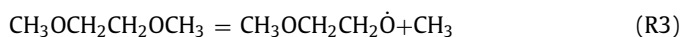
<sup>d</sup> CCSD(T)-F12/cc-pVTZ-F12//M06-2X/cc-pVTZ [8].

<sup>e</sup> CCSD(T)/cc-pV(D + T)Z//B2PLYPD3/6-311++g(d,p), this work.

coefficients for both C–O fission channels and pointed out that R1 is absolutely dominant over R2, through combined shock tube experiments and master equation analyses. These rate coefficients [16] are adopted in the current kinetic model.



Similarly, 1,2-DME may also undergo two different types of C–O bond fissions, respectively yielding CH<sub>3</sub> and CH<sub>3</sub>O (R3 and R4). The potential energy surface of 1,2-DME unimolecular decomposition was computed on the CCSD(T)-F12/cc-pVZT-F12/M06-2X/cc-pVTZ level of theory in our previous work [8]. The rate coefficients were obtained through an RRKM/master equation analysis, with the barrierless bond fission channels simply treated using the phase-space theory, since they are not competitive, compared to hydrogen abstraction reactions, under the low-pressure flame conditions where radicals are ubiquitous. The theoretical methods involved in Table 1 are expected to give energy values at similar accuracy. It is noted that the CH<sub>3</sub> producing C–O bond fissions in DMM and 1,2-DME require similar energies (around 82 kcal/mol); while a difference of about 3.5 kcal/mol is seen between the strengths of the other type of C–O bonds, whose cleaving results in CH<sub>3</sub>O, in the two fuel molecules. On such a basis, the same rate coefficient of R1 is estimated for R3, and that for R2, after a reduction of 3.5 kcal/mol in the activation energy, is used for R4. The breaking of the central C–C bond (R5) is an alternative decomposition channel of 1,2-DME, and it was actually considered as the exclusive pathway over the temperature range of 738–783 K in an early study [29], where 1,2-DME was used as a precursor to generate methoxymethyl radical (CH<sub>3</sub>OCH<sub>2</sub>). The rate coefficient for R5 is evaluated via an analogy to a similar bond fission in the central C–C of ethylene glycol (OHCH<sub>2</sub>–CH<sub>2</sub>OH), for which the pressure-dependent rate coefficients were computed based on the variable reaction coordinate-transition state theory (VRC-TST) in [30]. The strength of the C–C bond in OHCH<sub>2</sub>–CH<sub>2</sub>OH was computed to be 85.5 kcal/mol [30] at the QCISD(T)/CBS//B3LYP/6-311++G(d,p) level of theory, slightly higher than that of the C–C bond in 1,2-DME (81.8 kcal/mol [8]). So the activation energy is lowered by 3.5 kcal/mol to account for such a difference.

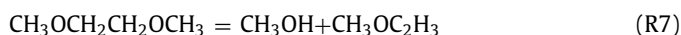
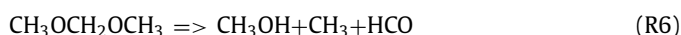


**Scheme 2.** Geometries for the transition states of the CH<sub>3</sub>OH elimination reaction of DMM obtained at B3LYP/6-311++g(d,p) level of theory.

Besides bond fissions, intramolecular elimination reactions producing alcohols are also possible unimolecular decomposition channels of ether fuels. Lizardo-Huerta et al. [31] discussed such reactions and reported corresponding rate rules. An irreversible lumped reaction producing methanol (CH<sub>3</sub>OH) (R6) was included in an early kinetic model [32] without a clear origin. However, a recent work by Golka et al. [16] indicated that no molecular channel was found in DMM decomposition. Possible molecular pathways are explored in this work, and the PES of DMM unimolecular decomposition, including both molecular elimination and bond fission channels, is provided Fig. S2 in the *Supplementary Material*. Two saddle points are found linking DMM with CH<sub>3</sub>OH and a biradical (CH<sub>3</sub>OCH and CH<sub>2</sub>OCH<sub>2</sub>) on the PES. Their geometries, with a three- and a five-membered cyclic structures, respectively, are presented in Scheme 2. The three-membered hydrogen transfer producing CH<sub>3</sub>OH is found as the lowest-lying pathway, with an energy barrier of 69.8 kcal/mol, slightly higher than those of the four-membered alcohol elimination reactions of diethyl ether (DEE) and dibutyl ether (DBE) reported in literature [31,33] (67.3–67.7 kcal/mol). Regarding the consumption of the simultaneously formed biradical CH<sub>3</sub>OCH, a recent work by Yu et al. [34] pointed out that the decomposition to CH<sub>3</sub>+HCO is energy favored, in comparison to other competing channels, including the dissociation into CH<sub>4</sub>+CO and the isomerization reactions forming acetaldehyde and oxirane. On such basis, the unstable CH<sub>3</sub>OCH is expected to rapidly decay into CH<sub>3</sub>+HCO, and a lumped reaction R6 is used in the current model to represent the CH<sub>3</sub>OH elimination reaction of DMM. The rate rule for secondary H transfer [31] is used for R6 with a multiplication of 4 by considering the number of transferable H atoms and the symmetrical structure of DMM. 1,2-DME can undergo a four-membered H transfer, eliminating a molecule of CH<sub>3</sub>OH and producing a methyl vinyl ether (CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub>) molecule



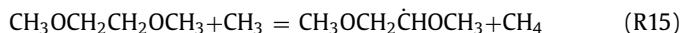
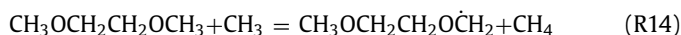
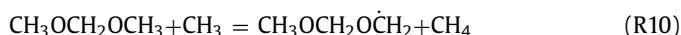
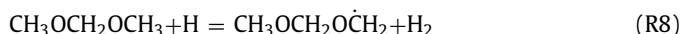
at the same time. The corresponding rate coefficient from our previous work [8] is used.



Rate coefficients for 1,2-DME bond fissions are mainly determined via analogies with similar reactions in the current kinetic model. More comprehensive investigations involving appropriate treatments for the multiple barrierless channels are needed in the future work to obtain more accurate kinetic parameters and their pressure-dependence.

### 3.3. Hydrogen abstraction reactions and radical dissociation reactions

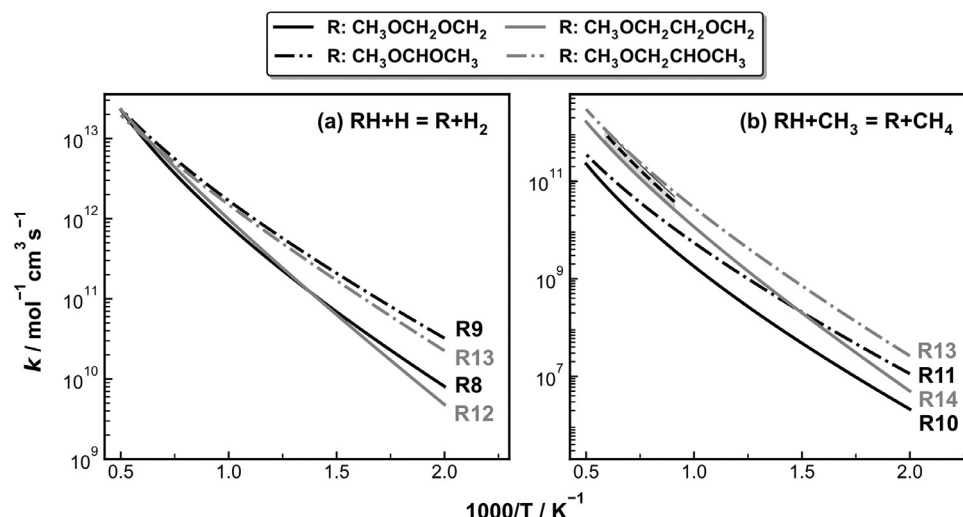
Hydrogen abstraction reactions by H and  $\text{CH}_3$  may significantly contribute to DMM and 1,2-DME consumption under pyrolysis conditions. Both DMM and 1,2-DME have two types of H atoms in their molecules. It is noted from Table 1 that the C–H bonds in the terminal ( $\text{CH}_3\text{O}-$ ) moieties of DMM and 1,2-DME have similar strengths, with BDEs being around 96 kcal/mol, which is also close to that of the C–H bond in dimethyl ether (DME) (96.1 kcal/mol [35]). Different moieties, ( $-\text{CH}_2-$ ) and ( $-\text{CH}_2\text{CH}_2-$ ), are embedded in the middle of DMM and 1,2-DME molecules, respectively. For DMM, the central C–H bond has a similar strength with the terminal one, while a difference of about 2 kcal/mol exists in the BDEs of the two types of C–H bonds in 1,2-DME (see Table 1). Rate coefficients of the hydrogen abstractions of DMM by H and  $\text{CH}_3$  (R8–R11) are reported in a theoretical work by Kopp et al. [28], based on a potential energy surface determined at the CCSD(T)/aug-cc-pV(D + T)Z//B2PLYPD3BJ/6–311++g(d,p) level of theory. The method used in the current work for 1,2-DME reactions (R12–R15), CCSD(T)/cc-pV(D + T)Z//B2PLYPD3/6–311++g(d,p), is expected to have similar accuracy. The two methods give close predictions for the energy barriers for hydrogen abstraction reactions of DMM by H, as can be seen from Table 1. The errors in the computed energy values are expected to be within 1 kcal/mol.



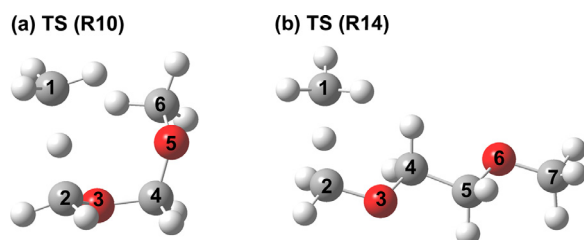
The computed energy barriers of 1,2-DME reactions (R12–R15) are listed in Table 1 and compared with those of the corresponding DMM reactions (R8–R11). The hydrogen abstractions at the terminal carbon atoms have similar energy barriers for both fuels, whereas the abstractions of the H atoms bonding to the internal carbon atoms are easier from 1,2-DME from the energy point of view, particularly for the abstractions by  $\text{CH}_3$ . The computed rate coefficients for hydrogen abstractions from 1,2-DME are shown in Fig. 1, and compared with those of DMM reactions reported by Kopp et al. [28].

The calculated results indicate that for both fuels, hydrogen abstractions from the terminal carbon atoms are relatively unfavored, and the abstractions by H atom from DMM and 1,2-DME have similar rate coefficients (see Fig. 1(a)). However, for hydrogen abstractions by  $\text{CH}_3$ , rate coefficients of the 1,2-DME reactions are remarkably higher than those of the DMM reactions (see Fig. 1(b)). This is unexpected, because corresponding reactions of the two fuels, particularly the abstractions of a-type hydrogen atoms by  $\text{CH}_3$  (R10 and R14), have similar energy barriers. The rate constant for the hydrogen abstraction by  $\text{CH}_3$  from dimethyl ether (DME) ( $\text{CH}_3\text{OCH}_3 + \text{CH}_3 = \text{CH}_3\dot{\text{O}}\text{CH}_2 + \text{CH}_4$ ), which is reported by Tranter et al. [36] by fitting their shock tube experiments, is also shown as a reference in Fig. 1(b). One may anticipate that hydrogen abstractions by the same agent from similar sites of different fuels, more specifically, from the terminal ( $\text{CH}_3\text{O}-$ ) structures of DME, DMM and 1,2-DME herein, to be very similar. Scheme 3 presents the optimized geometries of the transition states of R10 and R14. It is seen that the abstracting  $\text{CH}_3$  (C1) is close to the terminal methoxy group (O5–C6) of the DMM molecule, as the most stable cis-gauche (CG) conformer [37] of DMM has a spiral skeleton. This leads to a high barrier of the rotation around C4–O5 in TS(R10). Differently, the lowest-energy 1,2-DME conformer has a trans-trans-trans (TTT) structure [38], i.e. the C–O–C–O–C backbone is planar. The  $\text{CH}_3$  that attacks at the C–H bond has minor influences on the internal torsions in the 1,2-DME fragment. It is expected that the difference between the partition functions of the transition state and the reactants is more pronounced for R10 than for R14. This could account for the lower reaction rate of R10 though the C–H bonds in the methoxy groups of DMM and 1,2-DME have similar strengths. Eventually, for hydrogen abstractions from DMM by  $\text{CH}_3$ , the rate coefficients for reported by Vermeire et al. [13] were adopted in the current model to better capture the decomposition reactivity measured in this work as well as the JSR pyrolysis experiments in literature [13]. The chosen rate coefficients, which are based on theoretical calculations at CBS-QB3 level of theory [13], are about three times higher than those in [28], but the rate coefficient for R10 is still below those of R14 and the hydrogen abstraction from DME by  $\text{CH}_3$ .

Rate coefficients for the fuel radical  $\beta$ -scissions at 0.01, 1.0 and 100 atm are shown in Fig. S3 in the *Supplementary Material*. The a-type fuel radicals of DMM and 1,2-DME ( $\text{CH}_3\text{OCH}_2\dot{\text{O}}\text{CH}_2$  and  $\text{CH}_3\text{OCH}_2\text{CH}_2\dot{\text{O}}\text{CH}_2$ ), sharing the terminal ( $-\dot{\text{O}}\text{CH}_2$ ) moiety, decompose through the C–O  $\beta$ -scissions producing formaldehyde ( $\text{CH}_2\text{O}$ ). The two  $\beta$ -scission reactions have similar energy barriers (see Table 1), and the computed rate coefficients are also very close, with a less than 10% difference in their high temperature limits (HPLs) over the temperature range of 700–1200 K. Similarly, it was pointed out that the HPL of  $\text{CH}_3\text{OCH}_2\dot{\text{O}}\text{CH}_2$   $\beta$ -scission can be reproduced reasonably well with an analogy to the DME reaction  $\text{CH}_3\dot{\text{O}}\text{CH}_2 = \text{CH}_3 + \text{CH}_2\text{O}$  [28]. For the  $\beta$ -scissions of the b-type fuel radicals, the DMM fuel radical  $\text{CH}_3\dot{\text{O}}\text{CHOCH}_3$  can only dissociates into methyl formate ( $\text{CH}_3\text{OCHO}$ )+ $\text{CH}_3$ ; while two possible channels, separately leading to methyl vinyl ether ( $\text{CH}_3\text{OC}_2\text{H}_5$ ) and methoxy-acetaldehyde ( $\text{CH}_3\text{OCH}_2\text{CHO}$ ), compete for the consumption of the 1,2-DME fuel radical  $\text{CH}_3\text{OCH}_2\dot{\text{C}}\text{HOCH}_3$ . It is noteworthy from Fig. S3 that the rate coefficient for  $\text{CH}_3\dot{\text{O}}\text{CHOCH}_3 = \text{CH}_3\text{OCHO} + \text{CH}_3$  has a relatively weak temperature dependence. It is remarkably higher than those for the  $\beta$ -scissions of other fuel radicals over the entire investigated temperature range, particularly at temperatures below 1000 K. This is because the unique symmetrical methylenedioxy ( $-\text{OCH}_2\text{O}-$ ) structure in DMM brings two-fold effects: the energy barrier of  $\text{CH}_3\dot{\text{O}}\text{CHOCH}_3$   $\beta$ -scission (16.1 kcal/mol [28]) is much lower than those of the  $\text{CH}_3\text{OCH}_2\dot{\text{C}}\text{HOCH}_3$  dissociation channels (22.9 and 24.9 kcal); the reaction path degeneracy of  $\text{CH}_3\dot{\text{O}}\text{CHOCH}_3 = \text{CH}_3\text{OCHO} + \text{CH}_3$  is



**Fig. 1.** Rate coefficients for the hydrogen abstractions of DMM and 1,2-DME by (a) H and (b) CH<sub>3</sub>. The rate coefficients for DMM reactions were reported by Kopp et al. [28] and those for 1,2-DME reactions are calculated in this work. The dark dashed line in (b) is the rate coefficient for CH<sub>3</sub>OCH<sub>3</sub>+CH<sub>3</sub> = CH<sub>3</sub>OCH<sub>2</sub>+CH<sub>4</sub> reported by Tranter et al. [36] and the shaded area is the corresponding uncertainty.



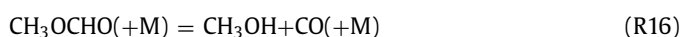
**Scheme 3.** Geometries of (a) the transition state for R10 optimized at the B2PLYPD3B/6-311 ++ g(d,p) level of theory [28] and (b) the transition state for R14 optimized at the B2PLYPD3/6-311++g(d,p) level of theory.

double of those for CH<sub>3</sub>OCH<sub>2</sub>CHOCH<sub>3</sub> = CH<sub>3</sub>OCH<sub>2</sub>CHO+CH<sub>3</sub> and CH<sub>3</sub>OCH<sub>2</sub>CHOCH<sub>3</sub> = CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub>+CH<sub>3</sub>O.

### 3.4. Consumption of important intermediates

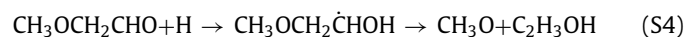
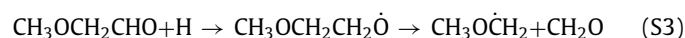
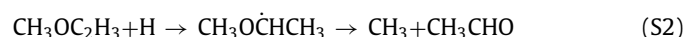
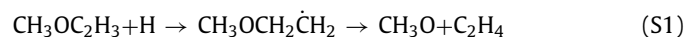
As discussed above, the production of b-type fuel radicals (CH<sub>3</sub>OCHOCH<sub>3</sub> and CH<sub>3</sub>OCH<sub>2</sub>CHOCH<sub>3</sub>) is favored over the a-type ones, and the subsequent β-scissions, especially for CH<sub>3</sub>OCHOCH<sub>3</sub>, proceed rapidly, resulting in fuel-specific molecular products, including CH<sub>3</sub>OCHO, CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub> and CH<sub>3</sub>OCH<sub>2</sub>CHO. These species are expected to be abundant in the pyrolysis of corresponding fuels, and their consumption potentially influence the speciation of other intermediates.

The decomposition of CH<sub>3</sub>OCHO through the molecular channel leading to CH<sub>3</sub>OH and CO (R16) is missing from the CH<sub>3</sub>OCHO sub-mechanism in AramcoMech 3.0 [26]. The rate coefficient reported by Felsmann et al. [39] was added to the current model. This theoretically determined rate coefficient agrees well with measurements in shock tube [40] and rapid compression machine [41] covering different temperature regimes.



Due to the unsaturated features of CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub> and CH<sub>3</sub>OCH<sub>2</sub>CHO, addition and addition-elimination channels (S1–S4) are considered in their consumption schemes. The inclusion of these channels are also inspired by the detection of the isomer pair of acetaldehyde (CH<sub>3</sub>CHO) and ethenol (C<sub>2</sub>H<sub>3</sub>OH). The CH<sub>3</sub>–C and C–OH structures do not exist in the fuel molecules,

and these species cannot be produced from the oxidation of C<sub>2</sub> hydrocarbons under the currently-concerned pyrolysis conditions, while the mentioned reactions (S1–S4) potentially account for the formation of CH<sub>3</sub>CHO and C<sub>2</sub>H<sub>3</sub>OH. The rate coefficients for the hydrogen addition reactions to the double C=C bond of CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub> were evaluated through analogies to the reactions H + C<sub>3</sub>H<sub>6</sub> → NC<sub>3</sub>H<sub>7</sub>/IC<sub>3</sub>H<sub>7</sub>. For hydrogen additions to the C–O bond in CH<sub>3</sub>OCH<sub>2</sub>CHO, rate expressions of the analogous reactions CH<sub>3</sub>CHO+H → C<sub>2</sub>H<sub>5</sub>O/C<sub>2</sub>H<sub>4</sub>OH are used in the current model.



Species thermochemical data are mainly from the core mechanism AramcoMech 3.0 [26], and for the fuel-related species that are not included in AramcoMech 3.0, the thermochemical data are from recent kinetic models [8,28]. The current kinetic model for DMM and 1,2-DME pyrolysis includes 606 species and 3182 reactions, for which the reaction mechanism and species thermochemical data are provided in the *Supplementary Material*. Simulations in this work are performed with the “PLUG” module of CHEMKIN-PRO [42] software. Reaction processes under different conditions specified by the corresponding measured temperature profiles were simulated. The resulting gas composition information at the ending point of each simulation was extracted to compare with the measurements obtained at the exit of the flow reactor.

## 4. Results and discussion

In this section, the predictive performances of the kinetic model, regarding the fuel reactivity and the speciation during DMM and 1,2-DME pyrolysis, are inspected with the measurements, ahead of further investigations comparing the two reaction systems. By combining experimental observations and relevant modeling analyses, crucial competing fuel consumption channels are identified. Common and fuel-specific reaction pathways accounting for the formation of separate species pools in DMM and 1,2-DME pyrolysis are illustrated.

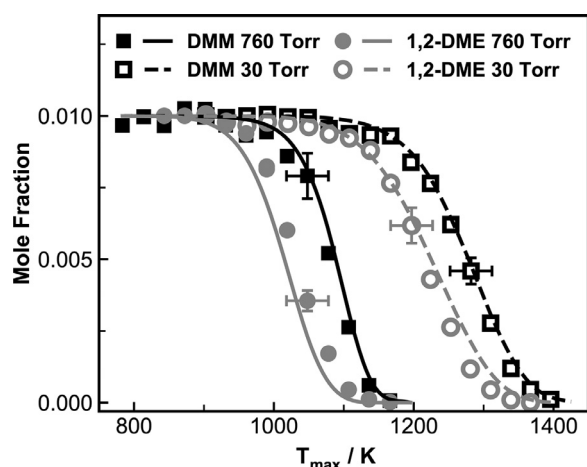


Fig. 2. Experimental (symbols) and simulated (lines) fuel mole fractions as a function of  $T_{\max}$  during DMM and 1,2-DME pyrolysis.

#### 4.1. The decomposition reactivity of DMM and 1,2-DME

Figure 2 shows the fuel (DMM and 1,2-DME) mole fractions as a function of  $T_{\max}$  at the two investigated pressures. The kinetic model can accurately capture the fuel conversion rates over the entire  $T_{\max}$  range, and it well predicts the observation that the consumption of 1,2-DME is faster than that of DMM under identical conditions. The higher decomposition reactivity of 1,2-DME can be partly rationalized by the fact that 1,2-DME can decompose through an additional C–C bond fission (R5) and hydrogen abstractions from 1,2-DME proceed at higher rates, as already discussed in the modeling section. To figure out the major fuel consumption pathways and how they evolve with the temperature conditions, rate-of-production (ROP) analyses are performed for both fuels throughout the  $T_{\max}$  windows of their consumption. The percentage contributions of different fuel consumption pathways are computed with the integrated ROP coefficients along the flow tube, and the  $T_{\max}$  dependent results for DMM and 1,2-DME are shown in Figs. 3 and 4, respectively.

Generally, for the consumption of both fuels at 760 Torr, the contribution of hydrogen abstraction reactions prevails over that of unimolecular decompositions, whose effects nevertheless enhance when raising the temperature. The C–O bond fissions leading to  $\text{CH}_3$  (R1) contributes significantly, reaching over one third at elevated  $T_{\max}$ s, to DMM consumption, while the equivalent C–O bond fission of 1,2-DME (R3) consumes a limited amount (less than 10%) of the fuel under all investigated temperature conditions at 760 Torr. In both DMM and 1,2-DME pyrolysis at 760 Torr, the fuel consumption is barely influenced by the C–O bond fissions leading to  $\text{CH}_3\text{O}$  (R2 and R4) and the  $\text{CH}_3\text{OH}$  elimination reactions (R6 and R7). The C–C bond fission (R5) is identified as the most important unimolecular decomposition channel of 1,2-DME, comparable with the hydrogen abstraction reactions (R12, R13, R15). The situations are different at 30 Torr, as the unimolecular channels (the bond fissions R1 and R5) become dominant fuel consumption channels in corresponding cases. In pyrolysis studies of diverse fuels [22–24,43] with the currently employed flow reactor configuration, it was consistently noted that unimolecular decompositions play a more important role in fuel consumption when decreasing the pressure. However, it is noteworthy that such a comparison is usually made when the same fraction of fuel is consumed at different pressures, for which the corresponding temperature distributions are distinct. Therefore, the varying trends of fuel consumption pathways are attributed not only to the pressure-dependent chemistry, but also to the different temperature conditions. Besides,

the scales of the residence time at 30 Torr and 760 Torr ( $10^{-3}$  s and  $10^{-1}$  s, respectively) differ by two orders of magnitudes, as mentioned previously.

Among the major DMM consumption pathways at 760 Torr, hydrogen abstractions by  $\text{CH}_3$  and by H have opposite dependence on  $T_{\max}$ . Hydrogen abstractions by  $\text{CH}_3$ , particularly from the central ( $-\text{CH}_2-$ ) moiety, dominate DMM consumption at relatively low  $T_{\max}$ s, while hydrogen abstractions by H have a greater contribution at elevated temperatures, and they intersect at  $T_{\max} = 1075$  K (see Fig. 3(a)). In the case of 1,2-DME, hydrogen abstractions by H (R12 and R13) consistently govern the fuel consumption over the investigated  $T_{\max}$ s at 760 Torr, though a decreasing trend can be spotted in their percentage contributions with the increase in  $T_{\max}$  (see Fig. 4(a)). The H and  $\text{CH}_3$  concentrations in the reactor can explain the different relative importance of hydrogen abstraction reactions in DMM and 1,2-DME pyrolysis. Figure 5 presents the simulated mole fraction profiles of fuel, H and  $\text{CH}_3$  along the flow reactor in DMM and 1,2-DME pyrolysis under specific temperature conditions where the fuel conversion rates are about 40%. Corresponding measured temperature profiles are also shown in Fig. 5. It is noted that in each case, fuel decomposition and radical formation quickly proceed in a reaction zone centered on the peak temperature. The reaction zone covers a distance range around 20–100 mm at 760 Torr, and spans over a wider regime at 30 Torr. The fuel consumption slows down beyond the reaction zone though the temperatures remain high. This is due to the insufficient radical concentrations, since both fuels are mainly consumed through hydrogen abstractions, as mentioned above. The radicals have much higher concentrations at 30 Torr, so that  $\text{CH}_3$  mole fractions at the exit of the flow reactor are measured as a function of  $T_{\max}$  in the current experiments. Figure 6 shows the  $\text{CH}_3$  mole fraction profiles in DMM and 1,2-DME pyrolysis, and the kinetic model overpredicts the measurements. However, for stable intermediates that are related to the consumption of  $\text{CH}_3$  consumption, such as methane ( $\text{CH}_4$ ) and ethane ( $\text{C}_2\text{H}_6$ ), the model can satisfactorily predict the measured profiles, as will be shown later. Thus, the discrepancy between the experimental and modeling  $\text{CH}_3$  mole fractions might be attributed to that a part of  $\text{CH}_3$  radicals quenches on the sampling nozzle, resulting in a loss in the measured mole fractions.  $\text{CH}_3$  has a higher abundance in DMM pyrolysis under identical conditions, though two ( $-\text{CH}_3$ ) moieties exist in both fuel molecules. The reasons lie in that the C–O bond fission leading to  $\text{CH}_3$  is more competitive in DMM consumption and that the rapid decomposition of the DMM fuel radical  $\text{CH}_3\text{O}\dot{\text{C}}\text{HOCH}_3$  produces a considerable amount of  $\text{CH}_3$ .

Sensitivity analyses for the fuels are performed to further understand which processes are the most important in the competition for the fuel decay and how they influence the decomposition reactivity of DMM and 1,2-DME. Specific temperature conditions, where the fuel conversion rates are about 40%, are chosen for the analyses in separate cases, and the resulting sensitivity spectra are shown in Fig. 7. At both pressures, the major chain initiation processes,  $\text{CH}_3\text{OCH}_2\text{OCH}_3 = \text{CH}_3\text{O}\dot{\text{C}}\text{H}_2 + \text{CH}_3$  (R1) and  $\text{CH}_3\text{OCH}_2\text{CH}_2\text{OCH}_3 = \text{CH}_3\text{O}\dot{\text{C}}\text{H}_2 + \text{CH}_3\text{O}\dot{\text{C}}\text{H}_2$  (R5), are the most significant channels promoting the consumption of DMM and 1,2-DME, respectively.  $\text{CH}_3$  is directly produced from R1, and also from the decomposition of  $\text{CH}_3\text{O}\dot{\text{C}}\text{H}_2$  subsequent to R5. Hydrogen abstraction reactions by  $\text{CH}_3$  are among the important reactions facilitating fuel consumption, especially at 30 Torr. In contrast, the  $\text{CH}_3$  self-recombination has the highest sensitivity inhibiting the fuel consumption in each case, as this reaction removes  $\text{CH}_3$  radicals from the reaction systems. However, one of the  $\text{CH}_3$  producing pathways, the 1,2-DME fuel radical  $\beta$ -scission,  $\text{CH}_3\text{CH}_2\dot{\text{C}}\text{HOCH}_3 = \text{CH}_3\text{OCH}_2\dot{\text{C}}\text{HO} + \text{CH}_3$ , hinders the fuel consumption. This is because the alternative channel of  $\text{CH}_3\text{CH}_2\dot{\text{C}}\text{HOCH}_3$

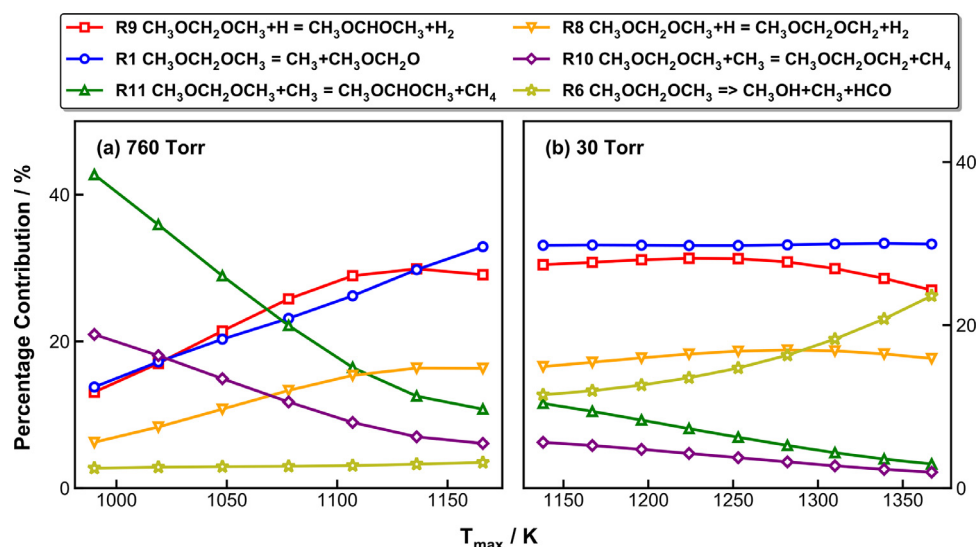


Fig. 3. Percentage contributions of the major fuel consumption pathways in DMM pyrolysis at (a) 760 Torr and (b) 30 Torr.

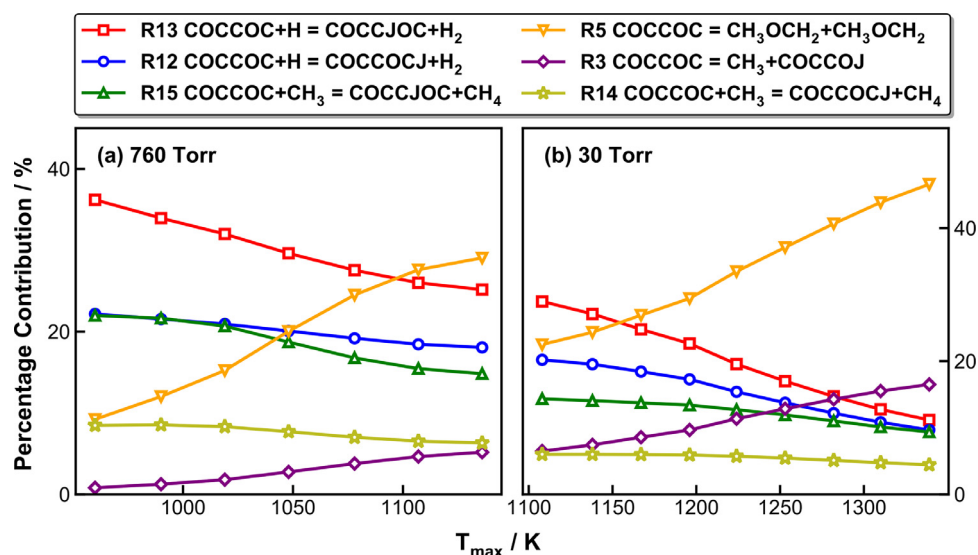


Fig. 4. Percentage contributions of the major fuel consumption pathways in 1,2-DME pyrolysis at (a) 760 Torr and (b) 30 Torr.

$\beta$ -scission,  $\text{CH}_3\text{CH}_2\dot{\text{C}}\text{HOCH}_3 = \text{CH}_3\text{OC}_2\text{H}_3 + \text{CH}_3\text{O}$ , yields  $\text{CH}_3\text{O}$  that decomposes into  $\text{CH}_2\text{O} + \text{H}$ , and H atoms enhances the system reactivity more efficiently than  $\text{CH}_3$ . H atoms and the main precursor radicals such as  $\text{CH}_3\text{O}$  and  $\text{HCO}$  cannot be directly produced from  $\beta$ -scissions of DMM fuel radicals. Therefore, hydrogen abstraction reactions by H atom play a more important role in 1,2-DME pyrolysis, compared to that of DMM pyrolysis, as can be seen in Fig. 7.

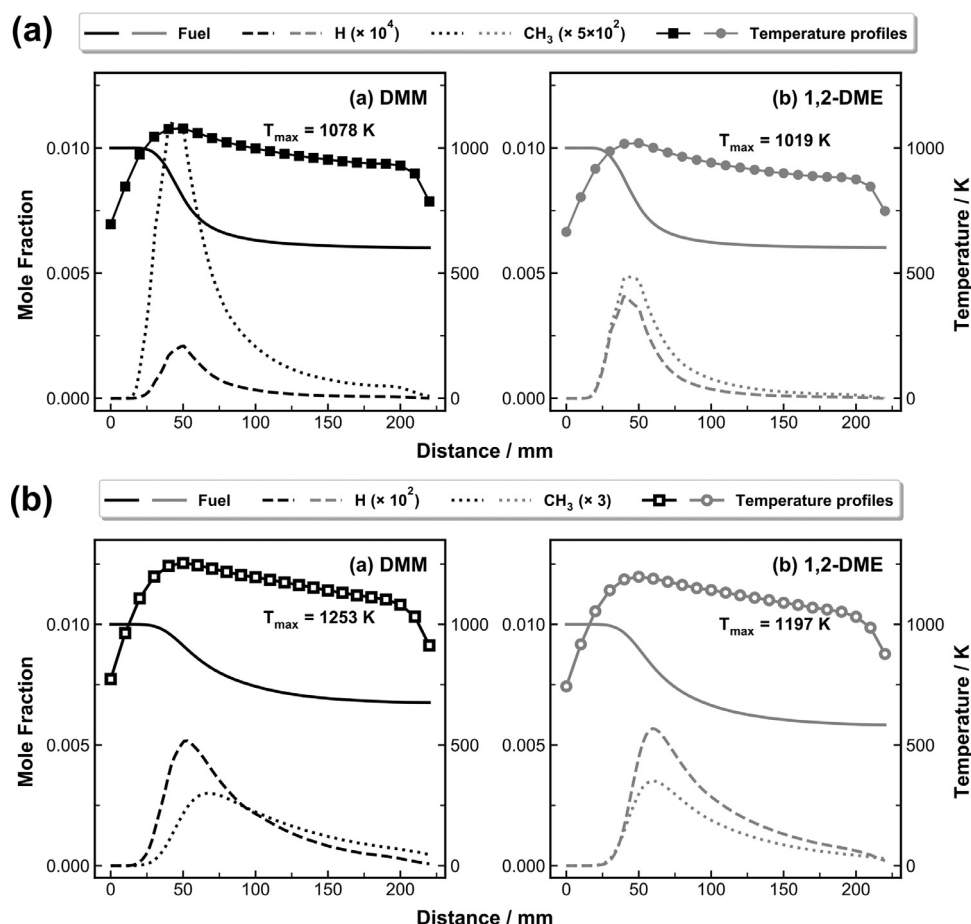
#### 4.2. Speciation in DMM and 1,2-DME pyrolysis

Mole fractions profiles of intermediates and products in DMM and 1,2-DME pyrolysis are presented as a function of  $T_{\max}$  in Figs. 8 and 9, respectively. The kinetic model proposed in this work accurately predicts the experimental data in the pyrolysis of both fuels. The DMM pyrolysis sub-mechanism in the current model is also tested with the species mole fraction profiles in the DMM JSR pyrolysis experiments reported in [13], as shown in Fig. S4 in the Supplementary Material. The current model can well predict the speciation during DMM pyrolysis under different experimental configurations. Among the studies relevant to

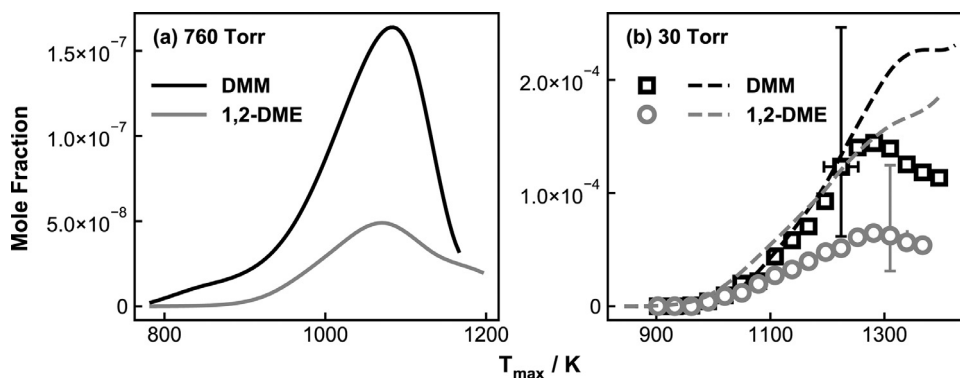
DMM pyrolysis, inconsistent conclusions were drawn about the relative importance of the two C–O bond fissions (R1 and R2) and the existence of the  $\text{CH}_3\text{OH}$  producing molecular channel (R6). These debated issues are worth being examined with the current experimental evidence and modeling analyses, before comparing the speciation behaviors and the responsible reaction schemes in DMM and 1,2-DME pyrolysis.

Only the overall rate of R1+R2 in DMM decomposition could be reliably derived by monitoring the H atom concentration history in shock tube studies [14,15], and corresponding modeling results are not affected by the branching ratio between R1 and R2 ( $k_{\text{R1}}/k_{\text{R2}}$ ). Discussions around  $k_{\text{R1}}/k_{\text{R2}}$  yielded different conclusions: Peukert et al. [14] noted that their measurements could be well reproduced if using a branching ratio reported in [13], where R2 was considered to govern the DMM unimolecular decomposition. However, according to Golka et al. [16], R1 clearly prevails over R2 in DMM thermal decomposition. The current detailed speciation measurements are used to inspect the influences of  $k_{\text{R1}}/k_{\text{R2}}$ .  $k_{\text{R1}}/k_{\text{R2}}$  is changed from 0 to 1 while fixing the total rate coefficient ( $k_{\text{R1}} + k_{\text{R2}}$ ), which is from [16] and adopted in the current model. The modeling results for all species mole frac-





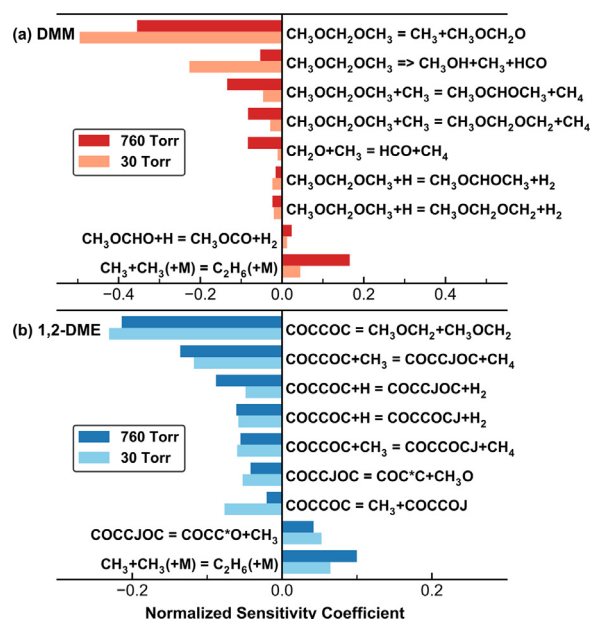
**Fig. 5.** Model-predicted mole fractions (lines) of fuel, H and CH<sub>3</sub> along the flow reactor in DMM and 1,2-DME pyrolysis under the temperature conditions where the final fuel conversion rates are around 40% at (a) 760 Torr and (b) 30 Torr. The measured temperature profiles are shown as the connected symbols.



**Fig. 6.** CH<sub>3</sub> mole fractions as a function of  $T_{max}$  in DMM and 1,2-DME pyrolysis at (a) 760 Torr and (b) 30 Torr. Symbols: measurements; lines: simulations.

tion profiles are shown in Fig. S5 in the *Supplementary Material*. In agreement with previous studies on DMM thermal decomposition [14,15], the fuel reactivity and the speciation of most intermediates are not obviously influenced by the variation of  $k_{R1}/k_{R2}$ . CH<sub>2</sub>O is the most impacted species, for which the measured and simulated mole fractions are presented in Fig. 10. When shifting the dominant DMM decomposition channel from R1 to R2, the CH<sub>2</sub>O mole fractions increase considerably, exceeding the upper limit of the measuring uncertainty. This is because the two fragments produced by R2, CH<sub>3</sub>O and CH<sub>3</sub>OCH<sub>2</sub>, both decompose producing CH<sub>2</sub>O. The CH<sub>2</sub>O mole fraction measurements in this work support the dominance of R1, as suggested by Golka et al. [16].

The CH<sub>3</sub>OH elimination channel (R6) is included in the current model, based on our findings on the DMM unimolecular decomposition PES, though the recent work by Golka et al. [16] suggests the absence of molecular channels in DMM decomposition. To assess the influences of R6, species mole fractions in DMM pyrolysis are simulated with R6 being excluded from the current model. The fuel consumption and CH<sub>3</sub>OH speciation are expected to be impacted, but according to the results presented in Fig. S6, obvious changes only show in the case at 30 Torr. ROP analyzed results indicate that CH<sub>3</sub>OH mainly comes from CH<sub>3</sub>OCHO decomposition through R16, while R6 only has a minor contribution, particularly at 760 Torr. At 30 Torr, the fuel decomposition reactivity is underestimated and the onset of CH<sub>3</sub>OH formation appears at a higher

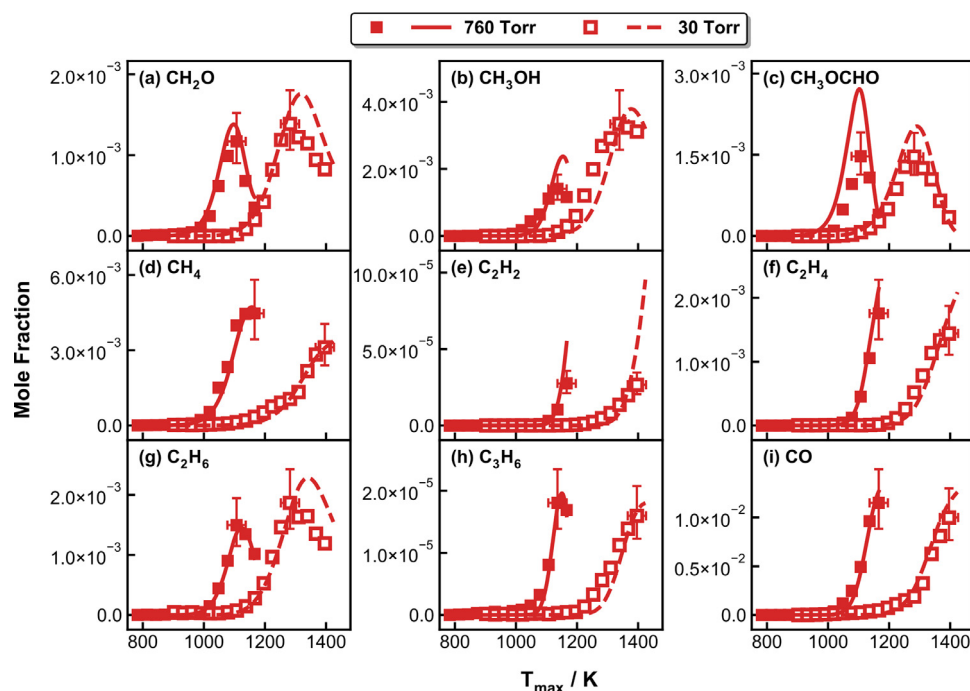


**Fig. 7.** Sensitivity analyzed results for (a) DMM at 760 Torr and  $T_{\max}=1078$  K, at 30 Torr and  $T_{\max} = 1253$  K; (b) 1,2-DME at 760 Torr and  $T_{\max}=1019$  K, at 30 Torr and  $T_{\max} = 1197$  K.

$T_{\max}$  if R6 is excluded. This indicates that the  $\text{CH}_3\text{OH}$  elimination reaction has a more obvious impact at 30 Torr. The above modeling analysis is based on the adoption of the  $k_{R1}/k_{R2}$  around 1 reported by Golka et al. [16]. Nevertheless, even if the real  $k_{R1}/k_{R2}$  is close to 0, the contribution of the  $\text{CH}_3\text{OH}$  elimination reaction (R6) to the fuel consumption still follows a similar trend as that shown in Fig. 3, i.e., being more significant at lower pressures. Besides, a higher fraction of  $\text{CH}_3\text{OH}$  is formed through R6 at 30 Torr than that at 760 Torr, consistent with the above discussion. Future experiments at low pressure conditions, joined by high-level theoretical investigations may better illustrate this molecular channel as well as its pressure dependence.

One major aim of this work is to compare the species pools of DMM and 1,2-DME pyrolysis, with regards to the species types and the different abundance of common intermediates. 1,2-DME pyrolysis results in a richer intermediates pool regarding the species types, compared to that of DMM pyrolysis, as can be judged from Figs. 8 and 9. Formaldehyde ( $\text{CH}_2\text{O}$ ) and methanol ( $\text{CH}_3\text{OH}$ ) are the common oxygenated intermediates in DMM and 1,2-DME pyrolysis. From the fuel molecular structural point of view, the terminal ( $\text{CH}_3\text{O}-$ ) moieties in both DMM and 1,2-DME may be released by losing and combining with an H atom, thus producing  $\text{CH}_2\text{O}$  and  $\text{CH}_3\text{OH}$ , respectively. Besides,  $\text{CH}_2\text{O}$  can also be formed from the ( $-\text{OCH}_2-$ ) structure in DMM and the ( $-\text{OCH}_2\text{CH}_2-$ ) structure in 1,2-DME. The  $\text{CH}_2\text{O}$  concentrations in 1,2-DME pyrolysis are three times higher than those in DMM pyrolysis under identical conditions. ROP analyses are performed for  $\text{CH}_2\text{O}$  formation in DMM and 1,2-DME pyrolysis at two conditions: 760 Torr,  $T_{\max} = 1078$  K and 30 Torr,  $T_{\max} = 1281$  K, where  $\text{CH}_2\text{O}$  concentrations are around the peak values in separate cases. The analyzed results along the flow tube are shown in Fig. 11.

Two reactions are shared in the  $\text{CH}_2\text{O}$  formation schemes in DMM and 1,2-DME pyrolysis:  $\text{CH}_3\text{OCH}_2 = \text{CH}_3 + \text{CH}_2\text{O}$  and  $\text{CH}_3\text{O} + \text{M} = \text{CH}_2\text{O} + \text{H} + \text{M}$ . These two reactions represent the  $\text{CH}_2\text{O}$  production from the central ( $-\text{OCH}_2-$ ) and terminal ( $\text{CH}_3\text{O}-$ ) moieties of the fuel molecules, respectively. The former reaction is the dominant pathway in all investigated cases, while the latter is only important in 1,2-DME pyrolysis, because  $\text{CH}_3\text{O}$  can be produced from the decomposition of the b-type fuel radical  $\text{CH}_3\text{OCH}_2\dot{\text{C}}\text{HOCH}_3$ , as discussed above. In both DMM and 1,2-DME pyrolysis,  $\beta$ -scissions of a-type fuel radicals,  $\text{CH}_3\text{OCH}_2\text{OCH}_2$  and  $\text{CH}_3\text{OCH}_2\text{CH}_2\text{OCH}_2$ , are important  $\text{CH}_2\text{O}$  contributors. As pointed out in the modeling section, the radicals  $\text{CH}_3\text{OCH}_2\text{OCH}_2$  and  $\text{CH}_3\text{OCH}_2$  decompose at similar rates, and  $\text{CH}_3\text{OCH}_2$  is mostly produced through the  $\beta$ -scission of  $\text{CH}_3\text{OCH}_2\text{OCH}_2$  in DMM pyrolysis. Consequently,  $\beta$ -scissions of  $\text{CH}_3\text{OCH}_2\text{OCH}_2$  and  $\text{CH}_3\text{OCH}_2$  have almost identical importance in  $\text{CH}_2\text{O}$  formation in DMM pyrolysis. On the other hand, the  $\beta$ -scission of  $\text{CH}_3\text{OCH}_2$  dominates the  $\text{CH}_2\text{O}$  formation in 1,2-DME pyrolysis, as a result of that a considerable amount of  $\text{CH}_3\text{OCH}_2$  radical is produced from the central



**Fig. 8.** Mole fraction profiles of intermediates and products in DMM pyrolysis. Symbols: measurements; lines: simulations.

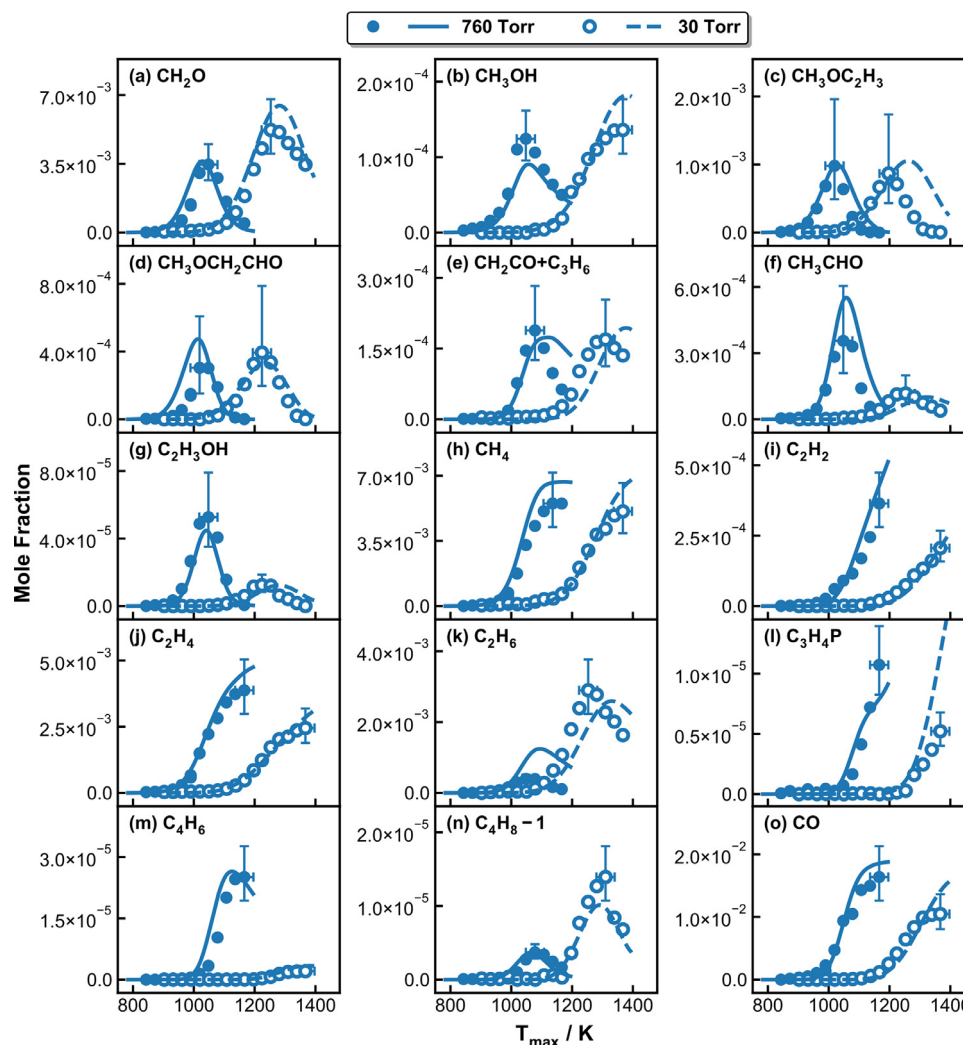


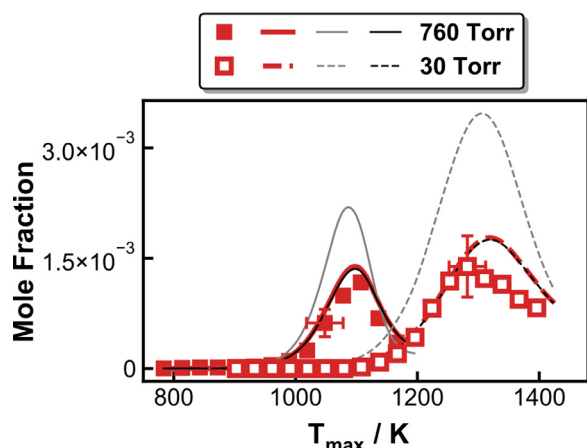
Fig. 9. Mole fraction profiles of intermediates and products in 1,2-DME pyrolysis. Symbols: measurements; lines: simulations.

C–C bond fission of 1,2-DME. Other fuel-specific  $\text{CH}_2\text{O}$  formation pathways with minor contributions are related to the consumption of crucial intermediates, specifically,  $\text{CH}_3\text{OCHO}$  in DMM pyrolysis and  $\text{CH}_3\text{OC}_2\text{H}_3$  in 1,2-DME pyrolysis.

The mechanisms responsible for  $\text{CH}_3\text{OH}$  formation are remarkably influenced by the pressures in both DMM and 1,2-DME pyrolysis. As mentioned previously, in DMM pyrolysis, the formation of  $\text{CH}_3\text{OH}$  is governed by the unimolecular decomposition of  $\text{CH}_3\text{OCHO}$  at 760 Torr, while the  $\text{CH}_3\text{OH}$  elimination reaction (R6) becomes similarly important at 30 Torr. Differently, in 1,2-DME pyrolysis at 760 Torr,  $\text{CH}_3\text{OH}$  is mostly converted from  $\text{CH}_3\text{O}$  by abstracting a H atom from the fuel or other intermediates, while at 30 Torr, the  $\text{CH}_3\text{OH}$  elimination from fuel molecule (R7) becomes the dominant channel. The unimolecular decomposition of the intermediate  $\text{CH}_3\text{OCH}_2\text{CHO}$  ( $\text{CH}_3\text{OCH}_2\text{CHO} = \text{CH}_3\text{OH} + \text{CH}_2\text{CO}$ ) has a minor contribution (a few percents) to  $\text{CH}_3\text{OH}$  production at both pressures.

Though both  $\text{CH}_2\text{O}$  and  $\text{CH}_3\text{OH}$  can be directly formed from fuel decomposition, their speciation is also impacted by the consumption of other intermediates, specifically,  $\text{CH}_3\text{OCHO}$  in DMM pyrolysis and  $\text{CH}_3\text{OC}_2\text{H}_3$  and  $\text{CH}_3\text{OCH}_2\text{CHO}$  in 1,2-DME pyrolysis. These species are fuel-specific and only observed in the pyrolysis of corresponding fuels (see Figs. 8 and 9). They mainly result from the  $\beta$ -scissions of the b-type fuel radicals  $\text{CH}_3\text{OCHOCH}_3$  and

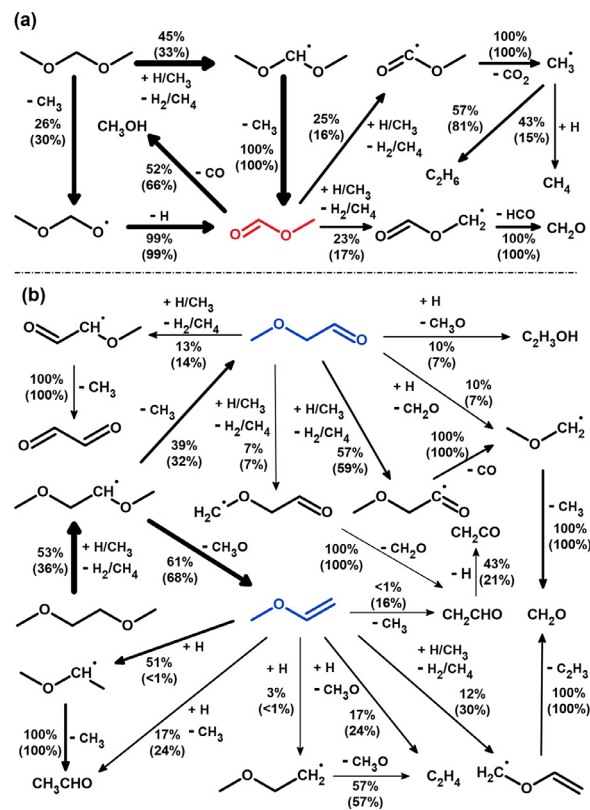
$\text{CH}_3\text{OCH}_2\dot{\text{C}}\text{HOCH}_3$ , which have distinct kinetic behaviors even under identical conditions. Figure 12 presents the formation and consumption pathways of the above-mentioned fuel-specific species, based on ROP analyses at specific temperature conditions where their concentrations are around the peak values. A great fraction of fuel carbon atoms flows into these intermediates, whose consumption adds to the diversity of the products pools in separate cases. For  $\text{CH}_3\text{OCHO}$  in DMM pyrolysis, besides the molecular channel R6 responsible for  $\text{CH}_3\text{OH}$  formation, the decomposition of the two  $\text{CH}_3\text{OCHO}$  radicals ( $\text{CH}_3\text{O}\dot{\text{C}}\text{O}$  and  $\dot{\text{C}}\text{H}_2\text{OCHO}$ ) promotes the formation of  $\text{CH}_3$  and  $\text{CH}_2\text{O}$ , respectively, and the  $\text{C}_2$  hydrocarbons originate from  $\text{CH}_3$  self-recombination. The reaction scheme in 1,2-DME pyrolysis is more complicated, as the  $\text{CH}_3\text{OC}_2\text{H}_3$  and  $\text{CH}_3\text{OCH}_2\text{CHO}$  formation channels compete in the consumption of  $\text{CH}_3\text{OCH}_2\dot{\text{C}}\text{HOCH}_3$ . The former is favored and with an increasing branching ratio at elevated temperatures, and the  $\text{CH}_3\text{OH}$  elimination reaction (R7) also has a minor contribution to  $\text{CH}_3\text{OC}_2\text{H}_3$  formation. Therefore,  $\text{CH}_3\text{OC}_2\text{H}_3$  is more abundant than  $\text{CH}_3\text{OCH}_2\text{CHO}$  in 1,2-DME pyrolysis (see Fig. 9(c) and (d)). Several  $\text{C}_2$  oxygenated intermediates are formed through the consumption of  $\text{CH}_3\text{OC}_2\text{H}_3$  and  $\text{CH}_3\text{OCH}_2\text{CHO}$ , as can be noted from Fig. 12(b). The isomer pair, acetaldehyde ( $\text{CH}_3\text{CHO}$ ) and ethenol ( $\text{C}_2\text{H}_3\text{OH}$ ), result from the addition or addition-elimination reactions of  $\text{CH}_3\text{OC}_2\text{H}_3$  and  $\text{CH}_3\text{OCH}_2\text{CHO}$ , respectively. The consumption of both  $\text{CH}_3\text{OC}_2\text{H}_3$



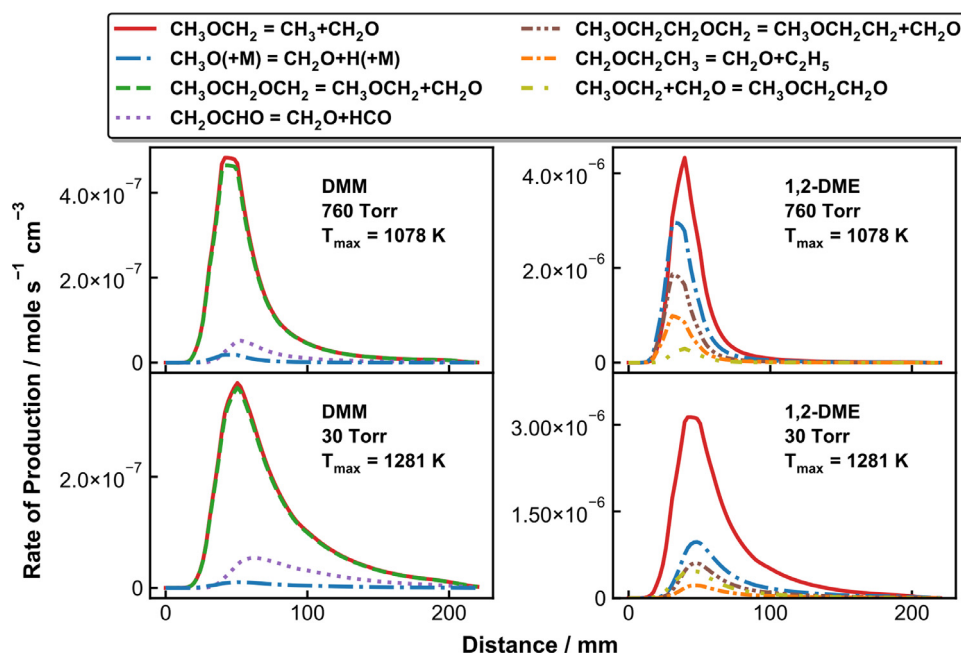
**Fig. 10.** Mole fraction profiles of  $\text{CH}_2\text{O}$  in DMM pyrolysis. Symbols: measurements; red lines: simulations with the current model; gray lines: simulations with the current model and  $k_{R1}/k_{R2}$  set at 0; black lines: simulations with the current model and  $k_{R1}/k_{R2}$  set at 1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and  $\text{CH}_3\text{OCH}_2\text{CHO}$  gives rise to  $\dot{\text{C}}\text{H}_2\text{CHO}$  radical, which further decomposes leading to ketene ( $\text{CH}_2\text{CO}$ ).

Figure 12(b) shows that  $\text{C}_2\text{H}_4$  and  $\text{C}_2\text{H}_3$  are released along the consumption pathways of 1,2-DME, whereas in DMM pyrolysis, the formation of  $\text{C}_2$  hydrocarbons only relies on the  $\text{CH}_3$  self-recombination. The onset  $T_{\text{max}}$ s and the relative abundance of  $\text{C}_2\text{H}_6$ ,  $\text{C}_2\text{H}_4$  and  $\text{C}_2\text{H}_2$  in DMM pyrolysis (see Fig. 8(e)–(g)) can prove that their formation follows the  $\text{CH}_3$  self-recombination and the subsequent stepwise dehydrogenation processes:  $\text{CH}_3 \rightarrow \text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_5 \rightarrow \text{C}_2\text{H}_4 \rightarrow \text{C}_2\text{H}_3 \rightarrow \text{C}_2\text{H}_2$ . In 1,2-DME pyrolysis, a similar (at 30 Torr) or even less (at 760 Torr) amount of  $\text{C}_2\text{H}_6$  is produced, compared to that in DMM pyrolysis, but  $\text{C}_2\text{H}_4$  and  $\text{C}_2\text{H}_2$  are more abundant. This indicates that unsaturated  $\text{C}_2$  hydrocarbons can be formed through alternative channels specific in 1,2-DME pyrolysis. Table 2 summarizes the formation pathways of the  $\text{C}_2$  hydrocarbons in 1,2-DME pyrolysis at 760 Torr and  $T_{\text{max}}$  of 1048 K and at 30 Torr and  $T_{\text{max}}$  of 1253 K based on ROP



**Fig. 12.** Reaction pathways for the formation and consumption of (a) methyl formate ( $\text{CH}_3\text{OCHO}$ ) in DMM pyrolysis at 760 Torr,  $T_{\text{max}} = 1107$  K and 30 Torr,  $T_{\text{max}} = 1281$  K and (b) methyl vinyl ether ( $\text{CH}_3\text{OCH=CH}_2$ ) and methoxy acetaldehyde ( $\text{CH}_3\text{OCH}_2\text{CHO}$ ) in 1,2-DME pyrolysis at 760 Torr,  $T_{\text{max}} = 1019$  K and 30 Torr,  $T_{\text{max}} = 1197$  K. The thickness of the arrows represents the carbon flux (originating from the fuel) through the corresponding reactions at 760 Torr. The numbers are percentage contributions to the consumption of the species on the source side of the arrows at 760 Torr and 30 Torr (in parenthesis).



**Fig. 11.** ROP analysis for formaldehyde ( $\text{CH}_2\text{O}$ ) along the flow reactor in DMM and 1,2-DME pyrolysis at 760 Torr,  $T_{\text{max}} = 1078$  K and at 30 Torr,  $T_{\text{max}} = 1281$  K.



**Table 2**

Formation pathways of C<sub>2</sub> hydrocarbons and their percentage contributions in 1,2-DME pyrolysis at 760 Torr and T<sub>max</sub> = 1048 K and at 30 Torr and T<sub>max</sub> = 1253 K. Reactions specific in 1,2-DME pyrolysis are highlighted in blue.

|                               | 760 Torr, T <sub>max</sub> = 1048 K                                                                                                                                                                                                                                                                                                                 | 30 Torr, T <sub>max</sub> = 1253 K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C <sub>2</sub> H <sub>6</sub> | CH <sub>3</sub> +CH <sub>3</sub> (+M) = C <sub>2</sub> H <sub>6</sub> (+M) (100%)                                                                                                                                                                                                                                                                   | CH <sub>3</sub> +CH <sub>3</sub> (+M) = C <sub>2</sub> H <sub>6</sub> (+M) (100%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| C <sub>2</sub> H <sub>5</sub> | <b>CH<sub>2</sub>OC<sub>2</sub>H<sub>5</sub> = CH<sub>2</sub>O+C<sub>2</sub>H<sub>5</sub> (81%)</b><br>C <sub>2</sub> H <sub>6</sub> +H = C <sub>2</sub> H <sub>5</sub> +H <sub>2</sub> (14%)<br>C <sub>2</sub> H <sub>6</sub> +CH <sub>3</sub> = C <sub>2</sub> H <sub>5</sub> +CH <sub>4</sub> (6%)                                               | <b>CH<sub>2</sub>OC<sub>2</sub>H<sub>5</sub> = CH<sub>2</sub>O+C<sub>2</sub>H<sub>5</sub> (64%)</b><br>C <sub>2</sub> H <sub>6</sub> +H = C <sub>2</sub> H <sub>5</sub> +H <sub>2</sub> (26%)<br>C <sub>2</sub> H <sub>6</sub> +CH <sub>3</sub> = C <sub>2</sub> H <sub>5</sub> +CH <sub>4</sub> (7%)<br>CH <sub>3</sub> +CH <sub>3</sub> = C <sub>2</sub> H <sub>5</sub> +H (2%)                                                                                                                                                                                                                                                                                                               |
| C <sub>2</sub> H <sub>4</sub> | C <sub>2</sub> H <sub>4</sub> +H(+M) = C <sub>2</sub> H <sub>5</sub> (+M) (45%)<br><b>CH<sub>3</sub>OCH<sub>2</sub>CH<sub>2</sub> = CH<sub>3</sub>O+C<sub>2</sub>H<sub>4</sub> (41%)</b><br><b>CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub>+H = CH<sub>3</sub>O+C<sub>2</sub>H<sub>4</sub> (9%)</b>                                                    | C <sub>2</sub> H <sub>4</sub> +H(+M) = C <sub>2</sub> H <sub>5</sub> (+M) (51%)<br><b>CH<sub>3</sub>OCH<sub>2</sub>CH<sub>2</sub> = CH<sub>3</sub>O+C<sub>2</sub>H<sub>4</sub> (45%)</b><br><b>CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub>+H = CH<sub>3</sub>O+C<sub>2</sub>H<sub>4</sub> (3%)</b>                                                                                                                                                                                                                                                                                                                                                                                                |
| C <sub>2</sub> H <sub>3</sub> | <b>CH<sub>2</sub>OC<sub>2</sub>H<sub>3</sub> = CH<sub>2</sub>O+C<sub>2</sub>H<sub>3</sub> (96%)</b><br>C <sub>2</sub> H <sub>3</sub> +CO = C <sub>2</sub> H <sub>3</sub> CO (4%)                                                                                                                                                                    | <b>CH<sub>2</sub>OC<sub>2</sub>H<sub>3</sub> = CH<sub>2</sub>O+C<sub>2</sub>H<sub>3</sub> (84%)</b><br>C <sub>2</sub> H <sub>4</sub> +H = C <sub>2</sub> H <sub>3</sub> +H <sub>2</sub> (10%)<br><b>CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub> = CH<sub>3</sub>O+C<sub>2</sub>H<sub>3</sub> (3%)</b><br>C <sub>2</sub> H <sub>4</sub> +CH <sub>3</sub> = C <sub>2</sub> H <sub>3</sub> +CH <sub>4</sub> (2%)<br>C <sub>2</sub> H <sub>3</sub> +H = C <sub>2</sub> H <sub>2</sub> +H <sub>2</sub> (39%)<br>C <sub>2</sub> H <sub>2</sub> +H(+M) = C <sub>2</sub> H <sub>3</sub> (+M) (27%)<br><b>CH<sub>3</sub>OC<sub>2</sub>H<sub>2</sub> = CH<sub>3</sub>O+C<sub>2</sub>H<sub>2</sub> (18%)</b> |
| C <sub>2</sub> H <sub>2</sub> | C <sub>2</sub> H <sub>2</sub> +H(+M) = C <sub>2</sub> H <sub>3</sub> (+M) (64%)<br><b>CH<sub>3</sub>OC<sub>2</sub>H<sub>2</sub> = CH<sub>3</sub>O+C<sub>2</sub>H<sub>2</sub> (17%)</b><br>C <sub>2</sub> H <sub>3</sub> +H = C <sub>2</sub> H <sub>2</sub> +H <sub>2</sub> (13%)<br>C <sub>2</sub> H <sub>2</sub> (+M) = H <sub>2</sub> CC(+M) (5%) | <b>CH<sub>3</sub>OC<sub>2</sub>H<sub>2</sub> = CH<sub>3</sub>O+C<sub>2</sub>H<sub>2</sub> (18%)</b><br>C <sub>2</sub> H <sub>2</sub> (+M) = H <sub>2</sub> CC(+M) (14%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

analyses. The chosen T<sub>max</sub>s are corresponding to the peak mole fractions of C<sub>2</sub>H<sub>6</sub> at two pressures. The above-mentioned CH<sub>3</sub> self-recombination still dominates C<sub>2</sub>H<sub>6</sub> formation, but fuel-specific reactions leading to unsaturated C<sub>2</sub> hydrocarbons, which are more efficient than the dehydrogenations, are introduced by CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub> consumption, particularly at 760 Torr and T<sub>max</sub> of 1048 K. The addition and addition-elimination reactions of CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub>+H account for almost half of the C<sub>2</sub>H<sub>4</sub> formation. Furthermore, the adduct CH<sub>3</sub>OCHCH<sub>3</sub> radical can isomerize to  $\dot{\text{C}}\text{H}_2\text{OCH}_2\text{CH}_3$ , which further decomposes into C<sub>2</sub>H<sub>5</sub>+CH<sub>2</sub>O. The  $\beta$ -scissions of CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub> fuel radicals,  $\dot{\text{C}}\text{H}_2\text{OC}_2\text{H}_3$  and CH<sub>3</sub>OCH $\dot{\text{C}}\text{H}$ , result in the formation of C<sub>2</sub>H<sub>3</sub> and C<sub>2</sub>H<sub>2</sub>, respectively. Due to the higher concentrations of C<sub>2</sub> molecules and radicals, larger hydrocarbons are formed through C<sub>1</sub>+C<sub>2</sub> and C<sub>2</sub>+C<sub>2</sub> recombination reactions in 1,2-DME pyrolysis. It is noteworthy that CH<sub>4</sub> mole fractions are higher in 1,2-DME pyrolysis though CH<sub>3</sub> is less sufficiently produced. This is because hydrogen abstractions by CH<sub>3</sub> leading to CH<sub>4</sub> are easier from 1,2-DME than from DMM, and the level of H atoms is higher in 1,2-DME pyrolysis so that there is a higher chance for CH<sub>3</sub> radicals to combine with H atoms to stabilize in CH<sub>4</sub>.

It is seen from the measured species pools that unsaturated hydrocarbons, particularly the soot precursors, are formed in limited amounts from DMM and 1,2-DME pyrolysis. This is thanks to the fact that the fuel carbon chains are fragmented by the embedded oxygen atoms. Carbonyl compounds, such as aldehydes, are formed from the dehydrogenation of (-CH<sub>2</sub>O-) or (CH<sub>3</sub>O-) units during the consumption of polyethers. This is different from the cases of the oxygenated fuels containing carbonyls, in which the C=O moieties tend to be directly released in CO or CO<sub>2</sub> [43–45], instead of participating in the formation of carbonyl intermediates. The thermal decomposition of specific oxygenated fuels, such as ethyl esters [46] and alcohols [21,47], may go through molecular elimination reactions with relatively low energy barriers, while for polyether fuels, bond fission reactions are the major unimolecular decomposition pathways under combustion relevant conditions.

## 5. Conclusions

A comparative study on the pyrolysis kinetics of DMM and 1,2-DME is carried out in this work. Pyrolysis experiments of both fuels are conducted with a flow tube reactor facility coupled to the molecular beam sampling photoionization mass spectrometry technique at two different pressures of 760 Torr and 30 Torr. Species mole fractions are measured at the exit of the flow reactor as a function of the temperature conditions inside the reactor.

Similar and unique reaction channels of DMM and 1,2-DME decomposition, which are related to their structural features, are compared. Based on that, a kinetic model is proposed, and it successfully captures the fuel reactivity and speciation in all investigated cases. Compared to DMM, 1,2-DME starts to decompose at lower temperatures, because the central (-CH<sub>2</sub>CH<sub>2</sub>-) moiety gives rise to a unique C–C bond fission channel, and hydrogen abstractions by CH<sub>3</sub> are relatively easier from 1,2-DME. The dominant fuel consumption channels in both DMM and 1,2-DME pyrolysis are strongly dependent on the pressure conditions. Hydrogen abstractions and bond fissions govern the fuel consumption at 760 Torr and 30 Torr, respectively. The CH<sub>3</sub>OH elimination reactions are considered in the current model, but they are not competitive enough to remarkably influence the fuel reactivity and the speciation under the currently investigated conditions. The measured species pools of DMM and 1,2-DME pyrolysis are distinct regarding the species types and abundance. CH<sub>2</sub>O is abundant in both reaction systems, as it can be released from the terminal (CH<sub>3</sub>O-) moieties as well as the central (-OCH<sub>2</sub>O-) or (-OCH<sub>2</sub>CH<sub>2</sub>O-) structures in fuel molecules. Fuel-specific species, including CH<sub>3</sub>OCHO in DMM pyrolysis and CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub> and CH<sub>3</sub>OCH<sub>2</sub>CHO in 1,2-DME, mainly originate from the  $\beta$ -scissions of the b-type fuel radicals, and their further consumption results in other intermediates. The abundant CH<sub>3</sub>OH in DMM pyrolysis predominantly results from CH<sub>3</sub>OCHO unimolecular decomposition. The addition-elimination reactions of CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub> and CH<sub>3</sub>OCH<sub>2</sub>CHO lead to the isomer pair of CH<sub>3</sub>CHO and C<sub>2</sub>H<sub>3</sub>OH in 1,2-DME pyrolysis. Besides, the decomposition of CH<sub>3</sub>OC<sub>2</sub>H<sub>3</sub> also produces unsaturated C<sub>2</sub> hydrocarbons, which accounts for the obviously higher contents of C<sub>3</sub>–C<sub>4</sub> hydrocarbons in 1,2-DME pyrolysis.

## Declaration of Competing Interest

None.

## Acknowledgments

This work is supported by the National Natural Science Foundation of China (Nos. 91841301 and U1832192). The authors gratefully acknowledge National Synchrotron Radiation Laboratory (NSRL) of China for providing beam time of the experiments.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2020.12.013](https://doi.org/10.1016/j.combustflame.2020.12.013).

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